Week 05: Joining related tables

Learning outcomes: CLO3

Lesson

- `INNER JOIN`: only matching rows. `LEFT JOIN`: all left rows, NULL where no match. Join on keys in `ON`.
- Joining parent to children repeats the parent. After joining communes, `COUNT(*)` counts communes; `COUNT(DISTINCT district_pcode)` counts districts.
- **Self join:** a table joined to itself. Use `a.code < b.code` to avoid duplicate pairs.
- `UNION` removes duplicates; `UNION ALL` keeps them.
- Rows with no match: `LEFT JOIN ... WHERE right.key IS NULL`. Empty is a valid answer.

Worked example

Run `example.sql` (how (`setup/README.md`)), then answer in `work.sql`.

Tables for this week

Read `cambodia`, write `student_work`. All columns are `NOT NULL`. Samples are the first two rows by code.

`cambodia.province`

Column	Type	Key	Meaning
`province_pcode`	`CHAR(4)`	PK	Province code
`province_name`	`VARCHAR(100)`		Province name (English)
`area_km2`	`DECIMAL(12,1)`		Area in km², one decimal place

Sample rows:

province_pcode	province_name	area_km2
KH01	Banteay Meanchey	6145.9
KH02	Battambang	11869.0

`cambodia.district`

Column	Type	Key	Meaning
`district_pcode`	`CHAR(6)`	PK	District code
`district_name`	`VARCHAR(100)`		District name (English)
`area_km2`	`DECIMAL(12,1)`		Area in km², one decimal place
`province_pcode`	`CHAR(4)`	FK	Province code

Sample rows:

district_pcode	district_name	area_km2	province_pcode
KH0102	Mongkol Borei	807.2	KH01
KH0103	Phnum Srok	761.7	KH01

`cambodia.commune`

Column	Type	Key	Meaning
`commune_pcode`	`CHAR(8)`	PK	Commune code
`commune_name`	`VARCHAR(100)`		Commune name (English)
`area_km2`	`DECIMAL(12,1)`		Area in km², one decimal place
`district_pcode`	`CHAR(6)`	FK	District code

Sample rows:

commune_pcode	commune_name	area_km2	district_pcode
KH010201	Banteay Neang	52.0	KH0102
KH010202	Bat Trang	40.3	KH0102

Join keys: `district.province_pcode = province.province_pcode`; `commune.district_pcode = district.district_pcode`.

More: data guide (`data/README.md`).

Lab tasks

1. Show each district with its province name.
2. Join communes → districts → provinces, sorted by commune code.
3. LEFT JOIN provinces to districts and count districts per province.
4. Find districts with no province (LEFT JOIN + IS NULL). Report the result, even if empty.
5. Self join provinces to find equal areas; if none, explain how the query would show them. Compare UNION vs UNION ALL on the same SELECT.

Homework

1. Join all three tables and say what one result row represents.
2. Show how joining communes changes a district count; fix it with COUNT(DISTINCT district_pcode).

Capstone sprint

C5: Add two join queries. Note repeated data or queries your design made hard.

Submit

Answers in `work.sql`, reflection in `reflection.md`, output in `results/`. Commit and push. See SUBMISSIONS.md.